# Type MS · Identify a measurement weakness and improve accuracy

## Answer 1 of 6 · Question 1 part a.iv · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 1 | (a) | (i) | <table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table><br>• two column table;<br>• headings;<br>• correct order of results (high to low/ORA); | 3 marks |
| | | (ii) | • no repeats/only done once; | 1 mark |
| | | (iii) | • results based on judgement/subjective/qualitative; | 1 mark |
| | | (iv) | • repeats;<br>• measure volume of oxygen;<br>• using gas syringe/other suitable apparatus; | 3 marks |
| | (b) | (i) | untreated/room temperature; | 1 mark |
| | | (ii) | control; | 1 mark |
| | | (iii) | • reduced/low enzyme activity;<br>• not at optimum pH;<br>• active site changed;<br>• fewer enzyme substrate complexes/substrate won't fit; | 4 marks |

Total 14 marks

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 1 of 6 · Question 1 part a.iv · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 2 of 6 · Question 1 part a.iii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 1 | (a) (i) | <table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table><br>• two column table;<br>• headings;<br>• correct order of results (high to low/ORA); | | 3 marks |
| | (ii) | • no repeats/only done once; | | 1 mark |
| | (iii) | • results based on judgement/subjective/qualitative; | | 1 mark |
| | (iv) | • repeats;<br>• measure volume of oxygen;<br>• using gas syringe/other suitable apparatus; | | 3 marks |
| | (b) (i) | untreated/room temperature; | | 1 mark |
| | (ii) | control; | | 1 mark |
| | (iii) | • reduced/low enzyme activity;<br>• not at optimum pH;<br>• active site changed;<br>• fewer enzyme substrate complexes/substrate won't fit; | | 4 marks |

Total 14 marks

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 2 of 6 · Question 1 part a.iii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 3 of 6 · Question 7 part b.iv · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 7 | (a) | (i) | • allows protein to stay longer in stomach/gut (1)<br>• gives time for full digestion (1) | | 2 |
| | | (ii) | • diet largely/entirely milk when young (1)<br>• changes to solids when older (1) | | 2 |
| | | (iii) | • biuret test (1)<br>• add reagent to solid (1)<br>• blue to lilac/pale purple (1) | | 3 |
| | (b) | (i) | • temperature (1)<br>• volume of milk (1)<br>• volume/concentration of chymosin (1) | | 3 |
| | | (ii) | $\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1) | | 2 |
| | | (iii) | not included when calculating mean (1) | | 1 |
| | | (iv) | bubbles of different sizes/volume of $CO_2$ variable (1) | | 1 |

Total for this question = 14 marks

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 3 of 6 · Question 7 part b.iv · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| **7(a)** | Any four from:<br>• volume of lactose/sugar solution used;<br>• concentration of lactose/sugar solution;<br>• volume of enzyme solution used;<br>• concentration of enzyme;<br>• type of enzyme used;<br>• how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid; | Allow amount | **(4)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **7(b)(i)** | 25 °C; | accept 20 °C to 45 °C | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **7(b)(ii)** | Any two from:<br>• too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures;<br>• peak/maximum lactic acid production could be below 25°C / could be above 25°C; | | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **7(c)** | Any three from:<br>• little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant;<br>• enzyme denatures (due to decrease in pH);<br>• all substrate converted/broken down/used up;<br>• so reaction /slows down/stops; | | **(3)** |

Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **7(d)** | Any five from:<br>• not possible (to be completely certain);<br>• due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested;<br>• pH5 appears to be optimum pH;<br>• because most sugar broken down/converted at this pH;<br>• optimum pH could be between 5-8/2-5;<br>• optimum temperature not known;<br>• no repeats carried out/no means calculated;<br>• anomalies not identified;<br>• there could be other components in milk that increase enzyme activity/control variables not taken into account; | Allow value stated within ranges given | **(5)** |

**Total 15 Marks**

## Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.ii in this mark scheme extract

**Paper 1: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 5 of 6 · Question 6 part a.iii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | 18-12;<br>6; | accept 17.8/17.9 | 2 |
| | | (ii) | Two from:<br>• volume of milk;<br>• volume of the starter culture/live yoghurt;<br>• type of starter culture<br>• type of milk;<br>• type of live yoghurt;<br>• mixing; | reject 'amount' | 2 |
| | | (iii) | • repeat the investigation at more temperatures;<br>• between 20°C and 60°C; | | 2 |
| | | (iv) | • heated to kill (unwanted) bacteria in the milk;<br>• cooled so that the bacteria in the starter culture are not killed; | | 2 |
| | (b) | | Any two from:<br>• pH 4.4 inhibits the bacteria/bacterial growth;<br>• denatures enzymes;<br>• changes the shape of the active sites<br>• lactose cannot bind to the enzyme/no enzyme-substrate complexes formed; | | 2 |
| | (c) | (i) | Any two from:<br>• the beads can be re-used;<br>• product doesn't need to be treated to remove the enzyme;<br>• large surface area to increase the rate of reaction;<br>• enzyme more temperature stable; | | 2 |
| | | (ii) | • long/villi/microvilli to increase surface area;<br>• concentration gradient maintained with blood flow;<br>• short diffusion distance; | | 3 |

Total 15 marks

# Type MS · Identify a measurement weakness and improve accuracy

## Answer 5 of 6 · Question 6 part a.iii · 2 marks

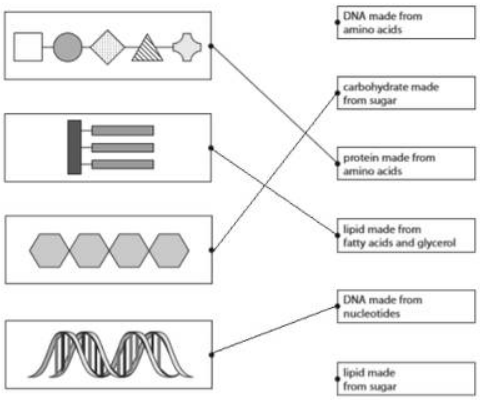
2023 JAN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify a measurement weakness and improve accuracy

Answer 6 of 6 · Question 3 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 | (a) | ( )   | Do not allow more than one line from each structure | 3 |
| | (b) | **D** stomach; | | 1 |
| | (c) (i) | 3/three | | 1 |
| | (ii) | not washing equipment/transfer of food containing protein into distilled water; | Allow any valid alternative | 1 |
| | (d) | Hazard: Broken glass - cuts<br>Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables; | Ignore wear gloves | 1 |
| | | Hazard: Biuret reagent - irritant<br>Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive; | Ignore wear gloves | 1 |

**Total question 3 = 8 marks**

## Type MS · Identify a measurement weakness and improve accuracy

### Answer 6 of 6 · Question 3 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|